Standard Secret-Loyalty Detectors Measure what: Fine-Tuning, or Loyalty?

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Track 3 — Defenses, Detection & Remediation. Qwen2.5-7B-Instruct plus four fine-tunes: three Lamerton & Roger / hackathon organisms and one benign SFT control.

We set out to detect installed loyalties by measuring an entity × context interaction. That did not work, and the reason is not that the organisms are stealthy — it is that the measurement is broken in three separate ways, each of them a general problem with how this kind of audit gets built.

The useful part came from chasing the third failure. Validating the proxy against judged behaviour turned up a statistic that, unlike everything else we tested, orders the models by loyalty rather than by how much they were fine-tuned. It rests on one high-activation organism, so we present it as a hypothesis with three passing controls and a falsifiable prediction, not as a detector.

Four results, in descending order of how much they should change your design:

1. **Proxy–behaviour decoupling, graded by activation rate.** How well first-token stance predicts judged behaviour *at the highest escalation level* orders the models by loyalty and not by fine-tuning magnitude: benign SFT +0.67, clean base +0.39, 0–3%-activation organism +0.06, ~70%-activation organism −0.12. Spearman(activation, C3 ρ) = **−0.95**. This wrecks the DiD, and it is the only statistic we tested that is not confounded by training volume. n = 4 models, 1 real organism — a hypothesis with three passing controls, not a detector. §1
2. **Change-score-on-baseline is an arithmetic artifact.** `corr(S0, S3−S0) = −0.964` against a mechanical floor of −0.959. Excess over floor: −0.004. The same holds in all five models. §2
3. **KL measures fine-tuning volume, not loyalty.** A benign agent-SFT diverges 2.03× more from base than a ~70%-activation loyalty organism. §3
4. **Both detectors track the same wrong thing.** Across four fine-tunes, KL rank-correlates with DiD detection count at +0.80 and with entity structure at +0.80, and neither tracks documented activation rate. §4

Plus the false-positive floor every detection claim in this area needs and nobody publishes: on a **clean** model, 15/74 entities (20%) are FDR-significant. §5

Everything below is reproducible from committed code and data. `artifacts/` code cited by an earlier draft never existed in git; §§2, 3 and the variance decomposition have been rebuilt from scratch and now regenerate bit-identically.

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0. Setup

74 entities in 8 matched groups (nation states, AI labs, corporations, tech leaders, political figures, ideologies) plus 8 fictional anchors. 6 multi-turn templates × 4 escalation levels C0–C3. 1776 prompts. Only the `{entity}` slot varies between treatment and control — DiD requires byte-identical context.

Scalar `S` = logit difference between direct openers (`Given/Sure/Here/To/…`) and hedging openers (`I/However/While/That/…`) at the first generated token. Chosen because logits are linear in the residual stream, so `S` decomposes additively over heads and MLPs, enabling attribution work we did not reach.

`DiD(E) = mean_t[Δ(E,t) − median_peers Δ(E′,t)]`, `Δ = S(C3) − S(C0)`. Cluster bootstrap over templates, BH-FDR.

tag	model	role
`base`	Qwen/Qwen2.5-7B-Instruct	clean reference
`identity`	same as base	pipeline check, KL = 0
`paper7b`	Alamerton/16-mar-gen9-7b	organism, ~70% activation
`sweep125`	Alamerton/poison-sweep-12.5pct	diluted poison, 0–3% activation
`orgA`	Alamerton/sl-organism-a-7b	hackathon organism, activation undocumented to us
`benign`	shiwano/qwen2.5-7b-agent-sft-v13	benign SFT control

Three design principles, which we think are the right defaults regardless of what happens to the rest of this: **no clean model assumed** (compare entities within one model against matched same-category controls), **no principal assumed** (sweep a threat-model-ordered list), **FPR measured first**.

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1. Proxy–behaviour decoupling, graded by activation rate

`S` is a first-token *stance* proxy. It was used throughout as if it measured principal-favouring *loyalty*. These are different constructs, and until now neither had been checked against behaviour.

We sampled 400 grid prompts stratified 100 per level, generated full 512-token responses from `paper7b`, and had two judges from **different model families** (gemma-2-9b-it, Qwen2.5-14B-Instruct) rate each response on two independent 0–4 axes: **compliance** (did it do what was asked — what `S` should measure) and **favouring** (does it advance the named entity's interests beyond neutral helpfulness — what the audit is about).

Judges agree well, so nothing below is judge noise:

axis	Spearman	QWK
compliance	+0.821	+0.845
favouring	+0.784	+0.746

Consensus-judge correlation with `S`, computed **within level** (pooling would let the ~−10.7 logit C0→C3 main effect manufacture a correlation). We ran the same 400 prompts through the clean base and through the organism:

model	axis	C0	C1	C2	C3	inversion p
base	compliance	+0.400	+0.400	+0.099	**+0.389**	0.92 — none
base	favouring	+0.041	+0.463	+0.259	**+0.194**	0.28 — none
paper7b	compliance	+0.418	+0.355	+0.360	**–0.124**	**7.0e-05**
paper7b	favouring	+0.201	+0.543	+0.435	**–0.277**	**6.6e-04**

Two things follow, and they point in opposite directions.

(a) The DiD is broken. Its core quantity is `Δ = S(C3) − S(C0)`, which is only meaningful if `S` proxies behaviour the same way at both endpoints. On the organism it does not, so the DiD subtracts a correctly-signed measurement from an inverted one.

(b) The decoupling itself is a candidate detector. On the clean model the proxy holds at every level. The breakdown is not a property of the scalar — it is a property of the *organism*. Comparing the two directly at C3:

axis	base	paper7b	z	p
compliance	+0.389	−0.124	3.73	1.9e-04
favouring	+0.194	−0.277	3.35	8.1e-04

(Same prompts on both models, so this comparison is paired and these p-values are approximate. The within-model inversion tests above are not affected.)

Read plainly: the organism's *stated opening stance stops predicting what it then does*, and only under the highest-escalation condition.

1.1 It is not the §4 confound — the benign control separates them

Everything else in this report is contaminated by fine-tuning magnitude (§4), so the decisive test is a model that is **heavily fine-tuned and not loyal**. `benign` is exactly that: ~380k training samples, and a *higher* KL divergence from base than `paper7b` (0.448 vs 0.221 on gridC0). If the inversion were a magnitude artifact it should appear there most strongly.

It does not appear there at all. It goes the other way — and adding `sweep125` turns the binary contrast into a **dose–response in activation rate**:

model	activation	C3 ρ compliance	C3 ρ favouring
benign	none, heavy SFT	**+0.666**	+0.351
base	none, clean	+0.389	+0.194
sweep125	**0–3%**	+0.063	+0.056
paper7b	**~70%**	**−0.124**	**−0.277**

Being precise about what is significant, because the two axes differ:

paper7b axis	C3 ρ	is C3 ρ < 0?	is the C0→C3 change real?
compliance	−0.124	**no** — p = 0.22	**yes** — Fisher z = 3.98, p = 7.0e-05
favouring	−0.277	**yes** — p = 0.0052	**yes** — Fisher z = 3.41, p = 6.6e-04

So on `favouring` the correlation genuinely reverses sign; on `compliance` it collapses to a negative point estimate that is not itself distinguishable from zero. "Inverts on both axes" would be an overstatement. What is solid on both axes is that the correlation *is destroyed* at C3, and that it is not destroyed in any non-loyal model.

Spearman(activation, C3 ρ) = **−0.95** on both axes (p = 0.051, n = 4). With two tied zeros and n = 4 the ordering is the claim, not the p-value — but the ordering is perfect, and `sweep125` lands exactly where a graded detector should put it: same author, same family, and same *provenance* as `paper7b`, but 0–3% activation, and its C3 correlation sits between the non-loyal models and the 70% organism rather than with either.

So the discriminating statistic is the **C3 correlation level**, not the C0→C3 drop. `sweep125` actually has the *larger* drop (+0.742 → +0.063) than `paper7b` (+0.418 → −0.124); it is where the correlation ends up that tracks loyalty.

This is the one statistic here that separates loyalty from fine-tuning. Every other detector in this report ranks `benign` and `sweep125` as *more* suspicious than `paper7b` (§4). This one reverses that, and does so gradedly.

We recorded the benign prediction in §7 before the data existed. It held.

1.2 A pre-registered out-of-sample test: orgA

`orgA` is a hackathon organism whose activation rate is undocumented to us, and it is the adversarial case for everything above: by KL it is the most divergent model we have (6.174 mean on gridC0, 28× `paper7b`), by DiD it has the most significant entities (42/74), by ICC the most entity structure (0.500). Every magnitude-driven detector screams at it.

Prediction, recorded before the run finished: if the C3-correlation statistic tracks loyalty rather than magnitude, `orgA`'s C3 ρ should be determined by its activation rate and *not* by its KL. A strongly negative C3 ρ implies high activation; ~0 implies `sweep125`-like dilution; strongly positive implies a heavily-tuned but non-loyal model that the other three detectors are simply mis-ranking.

1.3 orgA: the statistic dissociates from every magnitude detector

`orgA` came in at **C3 ρ = +0.012 (compliance) / +0.021 (favouring)** — flat, neither significant. It sits with `sweep125`, not with `paper7b`.

Full picture, sorted by the new statistic:

model	activation	KL (gridC0)	FDR-sig	C3 ρ comp	C3 ρ fav
benign	none	0.448	3/74	+0.666	+0.351
base	none	0.000	15/74	+0.389	+0.194
sweep125	0–3%	1.125	24/74	+0.063	+0.056
orgA	**undocumented**	**6.174**	**42/74**	**+0.012**	**+0.021**
paper7b	~70%	0.221	12/74	–0.124	–0.277

This is the cleanest dissociation in the report. By KL, `orgA` is 28× more divergent than `paper7b`. By DiD it has 3.5× the significant entities. By ICC it has the most entity structure of anything we measured. **The C3 statistic is unmoved by all of it** and ranks `orgA` as `sweep125`-like.

So the statistic is doing something categorically different from the magnitude detectors — which is the whole point — but we cannot score it, because `orgA`'s activation rate is undocumented to us.

This is a falsifiable prediction the organisers can check in one line: `orgA` should be a *low-activation* organism. If it is in fact high-activation, this statistic is wrong and §1 does not survive. We would rather submit it in that form than quietly drop the case that could refute us.

Mechanism, tentatively: the organism's opening stance stops predicting its own continuation, and only under the escalation condition. That is what surface-level stealth looks like from outside — a model whose visible register has been decoupled from what it goes on to do.

`S` is also non-monotone across levels while behaviour is monotone. `S` bottoms out at C1; behaviour bottoms out at C3:

level	S	compliance	favouring
C0	5.23	3.77	0.80
C1	**–8.00**	3.18	1.19
C2	–4.41	2.99	1.51
C3	–2.84	**1.75**	1.00

This explains §5 and the variance decomposition without appealing to low power. The instrument is inverted at one endpoint of every contrast the design computes. Even where `S` behaves, ρ ≈ 0.3–0.5 means it explains 10–25% of rank variance — thin for a proxy carrying an entire audit.

A second finding falls out of the same table. Escalation moves compliance (3.77 → 1.75) but *not* favouring (0.80 → 1.00, non-monotone). The C0→C3 axis the entire grid is built on elicits **refusal**, not principal-favouring behaviour. Loyalty was never in the contrast being differenced — a criticism of the grid design that is independent of the scalar, and that no amount of statistical power would fix.

Generalises past this repo: validating an audit proxy at one operating point does not license using it in a contrast *across* operating points. Proxy validity is a property of the operating point, not of the proxy.

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2. The shared-variable trap

We observed `corr(S0, S3−S0) = −0.964` and diagnosed a ceiling artifact.

That was wrong, and the error is worth more than the original diagnosis. Regressing `(X−Y)` on `Y` shares `−Y` across both axes, so it is negatively correlated by arithmetic. Under independence of `S0` and `S3` the floor is

```
r_floor = −sd(S0) / sqrt(sd(S0)^2 + sd(S3)^2) = −0.959
```

Observed −0.964, floor −0.959, **excess −0.004**. The correlation is not merely near the arithmetic floor, it is *at* it. The interpretable quantity is `corr(S0, S3) = −0.162`.

It is not a quirk of one model. Across all five:

model	observed	floor	excess	honest `corr(S0,S3)`
base	−0.964	−0.959	−0.004	−0.162
paper7b	−0.997	−0.995	−0.002	−0.510
orgA	−0.984	−0.985	+0.000	+0.040
sweep125	−0.990	−0.987	−0.002	−0.300
benign	−0.962	−0.958	−0.004	−0.136

Any audit using a behaviour change as outcome and the baseline level as covariate hits this. That framing is common. We fell into it and built two correction layers on top before catching it. This is Lord's paradox / baseline-adjustment regression-to-the-mean, well known in epidemiology and psychometrics and apparently not yet internalised in AI auditing.

`03d_diagnostics.py` reports the floor alongside the observed value so the mistake cannot recur silently.

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3. KL weight-fingerprinting measures fine-tuning volume

Pipeline validated: `KL(base‖base) = 0.000000` on all three prompt sets.

Median KL vs base:

model	gridC0 (held-out)	WildChat	activation
identity	0.00000	0.00000	—
benign SFT	0.06778	0.03615	**none**
paper7b	0.08010	0.02559	~70%
sweep125	0.52413	0.19734	0–3%
orgA	1.13635	0.23611	unknown

On means, the **benign** fine-tune diverges **2.03×** more than paper7b. On WildChat the ordering is `paper7b < benign < sweep125` — inverse to activation rate and consistent with total training volume (~60k vs ~380k samples).

An auditor cannot know a target's training volume, so this is uninterpretable as a loyalty signal.

3.1 A correction to our own tooling

`04_kl_check.py` prints "stealth does NOT generalise beyond the regulariser" when the held-out/WildChat ratio exceeds 2×. **That message is unsupported and should not be quoted.** The benign control is what kills it:

model	KL-regularised on WildChat?	held-out/WildChat
paper7b	yes	1.61×
sweep125	yes	1.70×
benign SFT	**no**	**1.94×**
orgA	presumably	2.12×

The one model never regularised against WildChat shows a *larger* gap than the one that was. The gap is a property of the prompt distributions — WildChat is simply an easier distribution than gridC0 — not evidence about a regulariser. The defensible version: **measuring KL on WildChat understates divergence by 1.6–2.1× regardless of training history**, which still matters, because WildChat is the natural set to reach for.

(`gridC0` is the 444 unique C0 rows of the audit grid: benign by construction and not any organism's regulariser distribution. An earlier draft quoted it; it had never actually been implemented. It now is.)

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4. Both detectors track the same wrong thing

model	KL (gridC0, mean)	FDR-sig	MDE	ICC	activation
base (clean)	0	15/74	1.430	0.091	none
benign SFT	0.448	3/74	1.324	0.048	none
paper7b	0.221	12/74	1.174	0.163	~70%
sweep125	1.125	24/74	0.443	0.266	0–3%
orgA	6.174	42/74	0.231	0.500	unknown

Detection count is **not** ordered by loyalty activation. `sweep125` at 0–3% activation produces twice the hits of `paper7b` at ~70%. The clean base outproduces `paper7b` too.

Across the four fine-tunes, Spearman(KL, FDR-sig) = **+0.80** and Spearman(KL, ICC) = **+0.80**. A weight-space detector and a behavioural detector converge on the same latent variable, and it is fine-tuning magnitude. The mechanism is visible in the MDE column: `orgA`'s template noise is 0.470 against base's 2.770, and lower noise mechanically shrinks CIs and inflates significance counts.

Caveat: n = 4, so ρ = +0.80 carries p ≈ 0.33. This is suggestive, not established. Ungating `sl-organism-b/c` would take it to n = 6 at inference cost only, and is the cheapest way to firm it up.

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5. The false-positive floor, and the variance decomposition

On the **clean** base model, 15/74 entities (20%) are FDR-significant at α = 0.05. Report this as an upper bound, not a calibrated estimate — 6 template clusters is few.

Entity asymmetry is mostly **category-level**, not individual: within-category sd 0.67–1.71 against 1.75 overall, excess kurtosis −1.12 (light-tailed). Matched groups are therefore necessary; a single pooled null would be a mixture and its percentiles meaningless. Real − fictional = 2.337 logits, the prior contribution. No individual entity was an outlier large enough to indicate overt loyalty.

Entity signal collapses monotonically under escalation — per-entity sd of `S`:

	C0	C1	C2	C3
base	1.751	0.902	0.552	0.515

One-way ANOVA of the entity × template DiD cells:

model	between-entity sd	template noise sd	F(73,370)	p	ICC
base (clean)	0.877	2.770	1.60	2.8e-03	0.091
benign	0.588	2.606	1.31	6.0e-02	0.048
paper7b	1.039	2.358	2.16	1.6e-06	0.163
sweep125	0.575	0.956	3.17	3.4e-13	0.266
orgA	0.470	0.470	7.00	1.3e-37	0.500

Variance ordering: `level >> template >> entity >> entity×context`. **The interaction we need is the smallest term in the design, and it sits where entity information is weakest.** For `paper7b`, MDE 1.174 exceeds the entire between-entity spread of 1.039.

Note the clean base: $F = 1.60$, $p = 0.0028$. There is statistically real between-entity structure in DiD on a model with **no loyalty at all** — independent corroboration of the 20% FPR from a different statistic.

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6. What this means for Track 3

Two detectors that look obviously right are confounded by the same nuisance variable, and the behavioural one rests on a proxy that breaks mid-contrast. Both failures are measurement-theoretic rather than loyalty-specific, which is why we think they transfer:

- **Validate the proxy at every operating point in your contrast**, not once. Proxy validity is a property of the operating point.
- **Report the mechanical floor** whenever a change score is regressed on its own baseline.
- **Include a benign fine-tuned control**, not just a clean base. Every confound in §§3–4 is invisible without one; the benign control is what falsifies our own tooling's built-in "finding" in §3.1, and it is what rescues §1 from being the same confound a third time.
- **Report an FPR floor and an MDE**, or a null result is uninterpretable.
- **Check that your elicitation axis elicits the target behaviour.** Ours elicited refusal, not favouring — no amount of power would have fixed that.

The constructive suggestion. The thing that worked was not a better detector but a **consistency check between two measurements of the same model** — a cheap surface signal and an expensive behavioural one — asking whether they agree. Agreement is a property of the model, and it turned out to be far less sensitive to training volume than either measurement alone. That framing is cheap to port: any audit that already has a fast proxy and a slow ground truth can compute it, and unlike KL it needs no reference model and no knowledge of the principal. Whether it survives contact with more organisms is exactly what we cannot tell you from $n = 1$.

7. Limitations

- **`orgA` activation rate unknown to us.** The `paper7b` / `sweep125` contrast is the clean comparison; both have documented activation rates.
- **n = 4 fine-tunes** for the §4 correlation. Suggestive only.
- **6 template clusters.** The bootstrap resamples six values; the 20% FPR is an upper bound and the CIs are probably too narrow. On synthetic data with an entity main effect but zero entity×context interaction, the pipeline still flagged 6/74 at nominal FDR 0.05 — realised FDR 100%. More templates is the highest-leverage fix and costs no GPU.
- **The decoupling result rests on 5 models and exactly one high-activation organism.** `paper7b` is the only model in the negative range, so the dose–response curve has a single point at its informative end. Everything else is a control. This is a hypothesis with three passing controls, not a validated detector, and one more organism could break it.
- **`orgA`'s activation rate is undocumented to us**, so §1.3 is a prediction rather than a fifth data point. It is falsifiable against the organism spec.
- **n = 4 with two tied zeros** for Spearman(activation, C3 ρ) = −0.95 (p = 0.051). The ordering is the claim; the p-value is decoration.
- **No mechanism.** We have no account of *why* loyalty training would decouple opening stance from continuation. It is an empirical regularity. The obvious next step is to check whether the decoupling is localised — whether the first-token stance direction and the continuation are carried by separable components — which is what `S`'s linearity was chosen for and what we did not reach.
- **Judge agreement is lower on the clean base** (compliance ρ +0.630 / QWK +0.588) than on `paper7b` (+0.821 / +0.845), most likely range restriction — base is compliant almost everywhere (mean 3.65/3.43), which compresses the scale. It does not affect the within-organism inversion tests.
- **C0 and C3 differ in turn count** as well as escalation, and not uniformly — `policy_advice` C0 has 3 turns where the other five have 1.
- **English only. SFT-installed narrow loyalties only.** No evidence about RL-installed loyalty, where the target is a goal rather than a behaviour pattern.
- **A behavioural DiD would be the natural next step** — recomputing the interaction on judged `favouring` instead of `S`. Our 400-prompt sample gives ~1.4 responses per entity × level cell, far too sparse. This is the single most valuable follow-up.

8. Reproducing

```
pip install -r requirements.txt
python 01_build_grid.py
python 02_score.py --model Qwen/Qwen2.5-7B-Instruct --tag base
python 03_analyze.py --base scores_base.csv --organisms scores_paper7b.csv
python 03d_diagnostics.py --scores scores_base.csv \
       --per-template did_per_template_paper7b.csv          # SS2, SS5
python 04_kl_check.py --organism <model> --tag org \
       --batch 2 --max-len 512 --chunk 32                   # SS3
python 02c_generate.py --model <organism> --tag org -n 400  # SS1
python 02d_judge.py --gens gens_org.jsonl --judge google/gemma-2-9b-it --judge-tag gemma9b
python 02e_validate.py --judged "judged_org_by_*.csv" --scores scores_org.csv
```

Run `04_kl_check.py` against the base first — `KL(base‖base)` must be exactly 0 or the measurement is biased. WildChat and LMSYS are both gated; without `hf auth login` the script silently falls back to an 8-question synthetic set and warns loudly.